

Two-Column Congruence Proofs Involving Triangles

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A **two-column proof** is an organized way of proving that a mathematical statement is true.

Statements and Reasons

The left column contains the **statements**, while the right column contains the corresponding **reasons**.

Every statement is paired with a reason that justifies it.

Starting a Proof

Begin with the **given information**, then use definitions, postulates, and theorems to justify each step.

Each new statement should follow logically from the ones before it.

Triangle Congruence Postulates

Common triangle congruence postulates are **SSS**, **SAS**, **ASA**, **AAS**, and the **HL Theorem** for right triangles.

These are usually the reason given in the final step of a congruence proof.

Using CPCTC

Once two triangles are proven congruent, use **CPCTC** — Corresponding Parts of Congruent Triangles are Congruent — to prove corresponding sides or angles are congruent.

CPCTC can only be used *after* congruence has already been proven.

Every Step Needs a Reason

Every statement in a proof must have a valid mathematical reason.

A statement with no reason — or the wrong reason — makes the proof incomplete.

Examples

Example 1 — SAS Proof

Given: $AB \cong DE, \angle A \cong \angle D, AC \cong DF$

Prove: $\triangle ABC \cong \triangle DEF$

Statement: $AB \cong DE, \angle A \cong \angle D, AC \cong DF$

Reason: Given

Statement: $\triangle ABC \cong \triangle DEF$

Reason: SAS

Example 2 — SSS Proof

Given: $AB \cong DE, BC \cong EF, AC \cong DF$

Prove: $\triangle ABC \cong \triangle DEF$

Statement: $AB \cong DE, BC \cong EF, AC \cong DF$

Reason: Given

Statement: $\triangle ABC \cong \triangle DEF$

Reason: SSS

Example 3 — Using CPCTC

Given: $\triangle ABC \cong \triangle DEF$

Prove: $\angle B \cong \angle E$

Statement: $\triangle ABC \cong \triangle DEF$

Reason: Given

Statement: $\angle B \cong \angle E$

Reason: CPCTC

Practice Exercises

Practice 1

In $\triangle ABC$ and $\triangle DEF$, $AB \cong DE$, $BC \cong EF$, and $\angle B \cong \angle E$. Construct a two-column proof that $\triangle ABC \cong \triangle DEF$.

Practice 2

Given $AB \cong DE$, $AC \cong DF$, and $BC \cong EF$. Construct a two-column proof that $\triangle ABC \cong \triangle DEF$.

Practice 3

Given $\triangle ABC \cong \triangle DEF$. Construct a two-column proof that $\angle C \cong \angle F$.

Practice 4

Given $AB \cong CD$, $\angle A \cong \angle C$, and $\angle B \cong \angle D$. Construct a two-column proof that $\triangle ABE \cong \triangle CDF$.

Practice 5

Given $\triangle ABC \cong \triangle DEF$. Construct a two-column proof that $\overline{AC} \cong \overline{DF}$.

Activities

Activity 1

Given $\angle M \cong \angle P$, $\angle N \cong \angle Q$, and $MN \cong PQ$. Construct a two-column proof that $\triangle MNO \cong \triangle PQR$.

Activity 2

In right triangles $\triangle RST$ and $\triangle ABC$, $\angle S$ and $\angle B$ are right angles, $RT \cong AC$, and $ST \cong BC$. Construct a two-column proof that the triangles are congruent.

Activity 3

Given $\triangle PQR \cong \triangle XYZ$. Construct a two-column proof that $QR \cong YZ$.

Activity 4

In $\triangle RST$, RS is the common side of $\triangle RSM$ and $\triangle RSN$. Given $RM \cong RN$ and $SM \cong SN$, construct a two-column proof that $\triangle RSM \cong \triangle RSN$.

Activity 5

Given $LM \parallel NO$, $\angle L \cong \angle N$, and $MN \cong LO$. Construct a two-column proof that $\triangle LMN \cong \triangle NOL$.

Thank you!